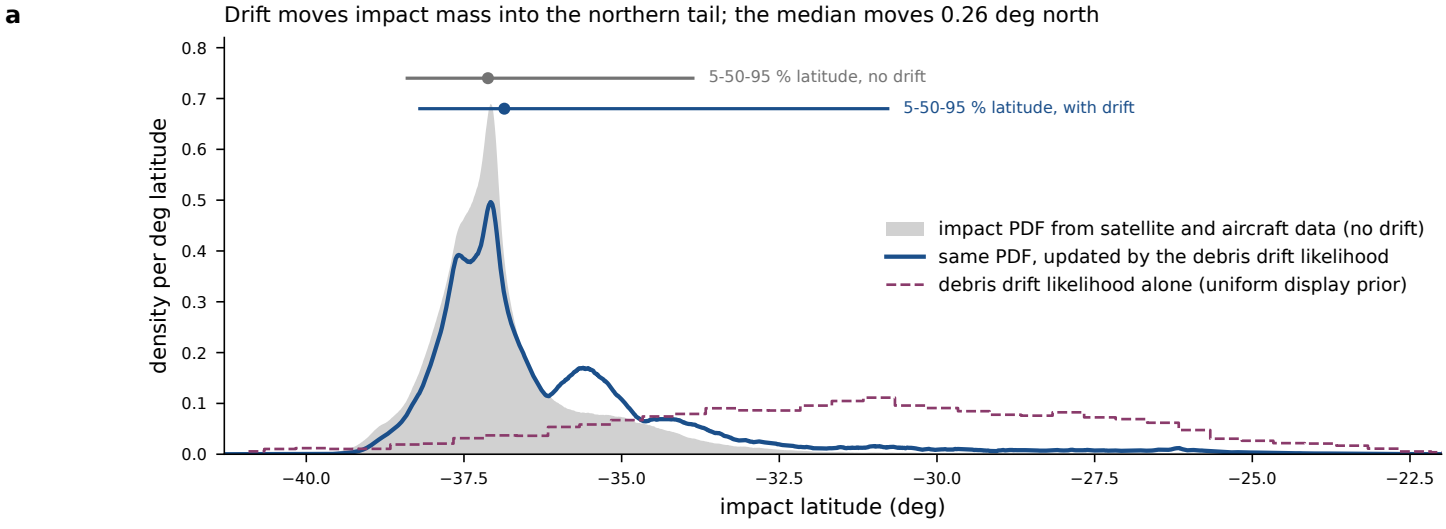
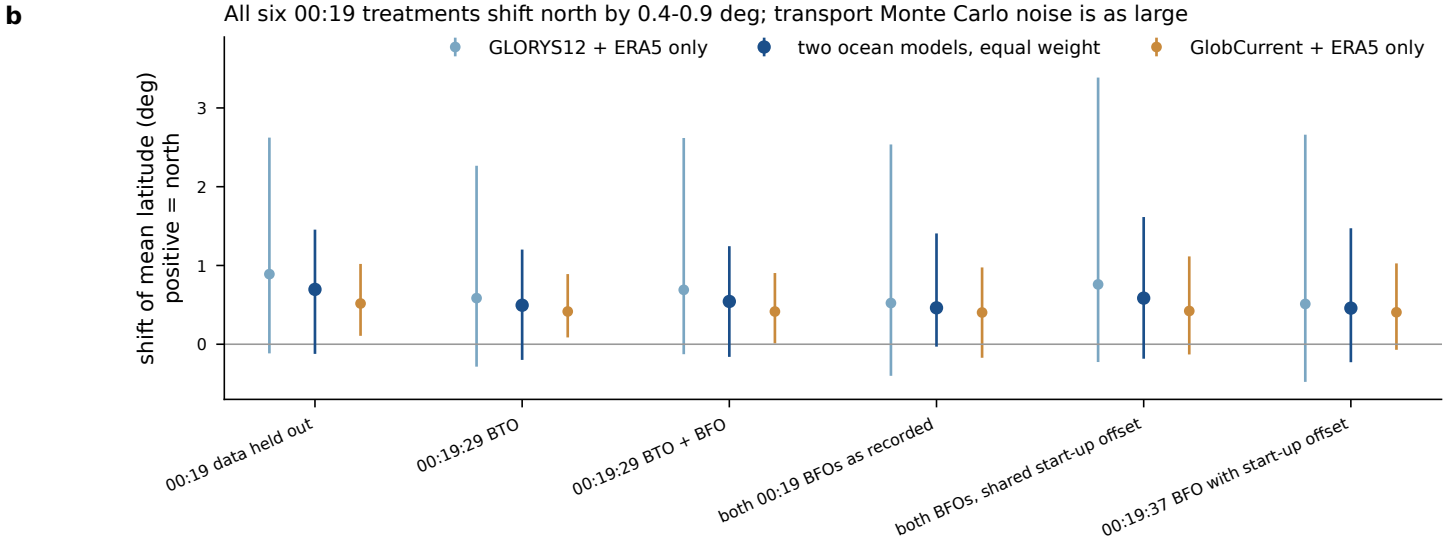
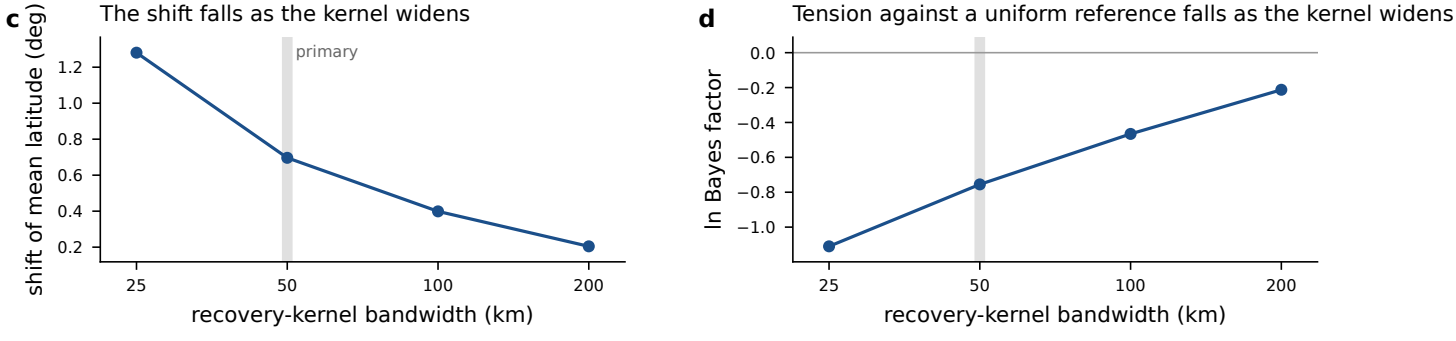






Plain English. The impact PDF comes from the satellite data and the aircraft performance model (run J, waypoints family, 24 seeds), continued to impact by the end-of-flight model. The 00:19 log-on has two possible causes: fuel exhaustion, or another cause (prior odds 1 to 3 against the other cause); under the other cause the aircraft is silent. The drift likelihood uses the nine securely identified debris finds and three types of object. It comes from forward drift from 710 start points, with two ocean models of equal weight. Error bars show the 95 % range of transport Monte Carlo noise. The absence of finds in Australia is not in this figure. The uncontrolled-descent physics is not calibrated.

Technical. Core run J (tag run-j, 1244b7c), waypoints stratum (family posterior 0.9982); end of flight run J (run-i-dive recipe, round-3 laws, bank limit U[52,90] deg); cells 'D total q=1/3 (C2 silent)', ln Z and ESS reproduce the end-of-flight table exactly. Drift: run A surfaces, GLORYS12V1 0.494 m + ERA5 and GlobCurrent my-p1d 0 m + ERA5 (windage offset U(-0.75,-0.60) % of 10 m wind, product-relative), 30 NM nodes, 1e5 particles per node, kernel 50 km except c-d; surfaces rebuilt 16 Oct from run-i-dive per-impact columns (max |d ln L| 1.7e-5). Rows not computed carried at the seed mean (<=0.03 % of weight). Noise: 200 mean-preserving log-normal node perturbations, sigma from split halves per 1-deg band; seed-bootstrap SD <=0.14 deg. Reference: uniform per unit area over fully evaluated drift cells. Gap G26 open; breakup rows at intact entry; uncontrolled arm uncalibrated (Boeing leave-one-out gate failed).